

FINITE-SIZE CORRECTIONS FOR 4-POINT PROBE MEASUREMENTS (rev 07Jan04)

With the probes centered on a very wide (lateral dimension $d \gg s$) and very thin (thickness $t \ll s$) sample, with s the probe spacing, the resistivity is given by:

$$\rho = \frac{\pi}{\ln(2)} t \left(\frac{V}{I} \right) = 4.5324 t \left(\frac{V}{I} \right)$$

For a sample of finite width and non-negligible thickness this should be multiplied by correction factors f_1 and f_2 :

$$\rho = \frac{\pi}{\ln(2)} t \left(\frac{V}{I} \right) f_1 f_2 = 4.5324 t \left(\frac{V}{I} \right) f_1 f_2$$

where

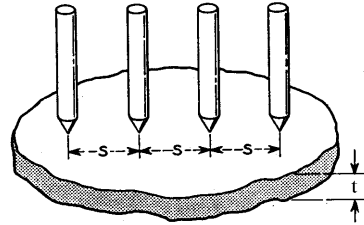
$f_1 = f_1(t/s)$ = finite thickness correction :

$f_1 = f_{11}(t/s)$ = for an insulating bottom boundary

$$f_{11} = \frac{\ln(2)}{\ln \left[\frac{\sinh(t/s)}{\sinh(t/2s)} \right]}$$

$f_1 = f_{12}(t/s)$ = for a conducting bottom boundary

$$f_{12} = \frac{\ln(2)}{\ln \left[\frac{\cosh(t/s)}{\cosh(t/2s)} \right]}$$



in our probe: $s=1.59$ mm

$$\text{for } t \ll s, \quad f_{11} \approx 1 \quad \text{and} \quad \rho \approx \frac{\pi}{\ln 2} t \left(\frac{V}{I} \right) f_2 = 4.5324 t \left(\frac{V}{I} \right) f_2$$

$$f_{12} \approx \frac{8}{3} \ln(2) \frac{s^2}{t^2} \quad \text{and} \quad \rho \approx \frac{8\pi}{3} \frac{s^2}{t} \left(\frac{V}{I} \right) f_2$$

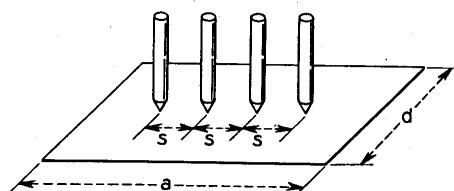
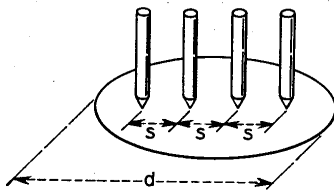
$$\text{for } t \gg s, \quad f_{11} = f_{12} \approx 2 \ln(2) \frac{s}{t} \quad \text{and} \quad \rho \approx 2\pi s \left(\frac{V}{I} \right) f_2$$

f_2 = finite width correction :

$f_2 = f_{2C}(d/s)$ = for a circular sample of diameter d

$f_2 = f_{2R}(a/d, d/s)$ = for a rectangular sample of width d and length a

for $d/s \gg 1$, $f_{2C} = f_{2R} \approx 1$



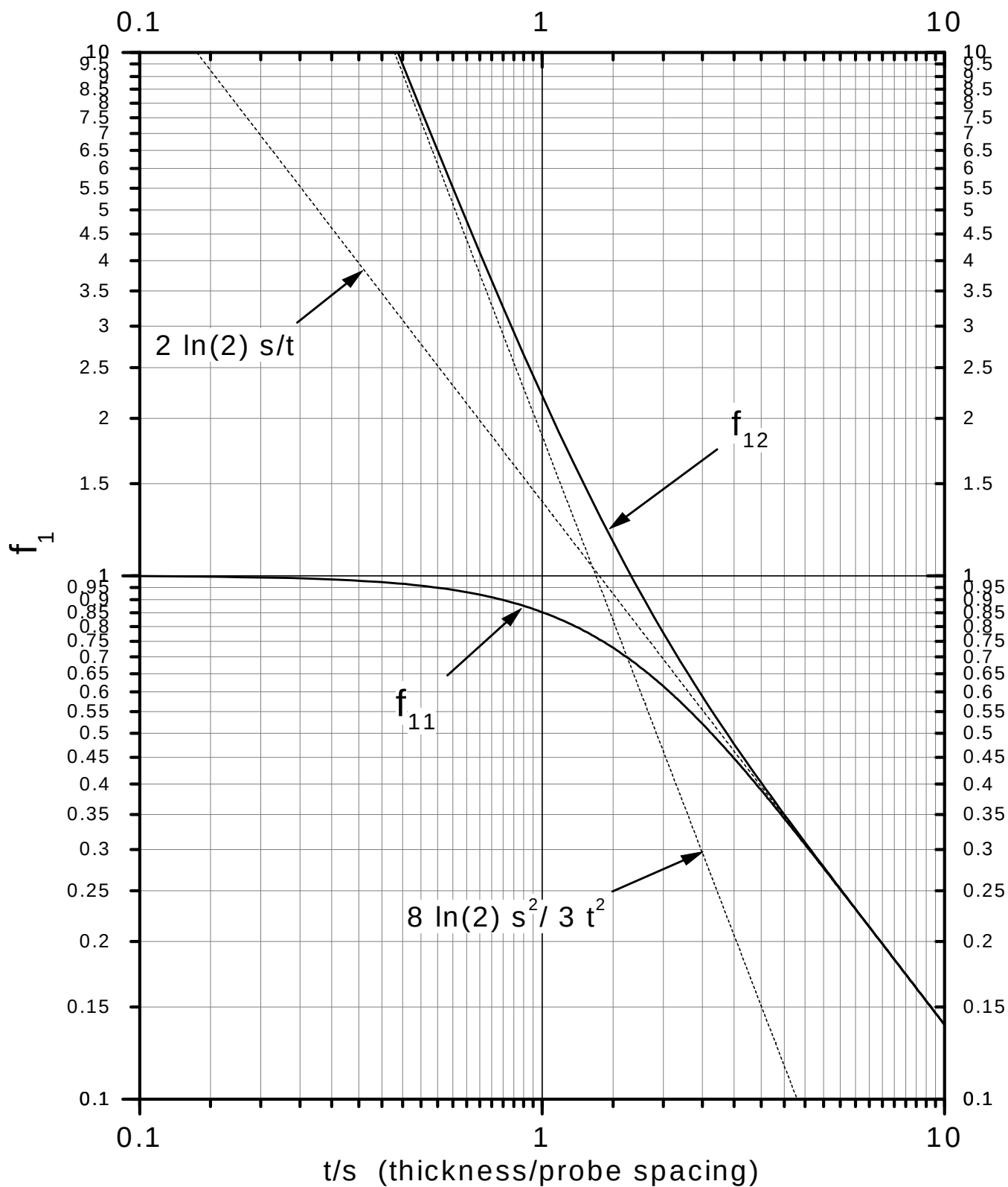
Original references:

A. Uhler Jr., The Bell System Technical Journal **34**, 105 (1955)

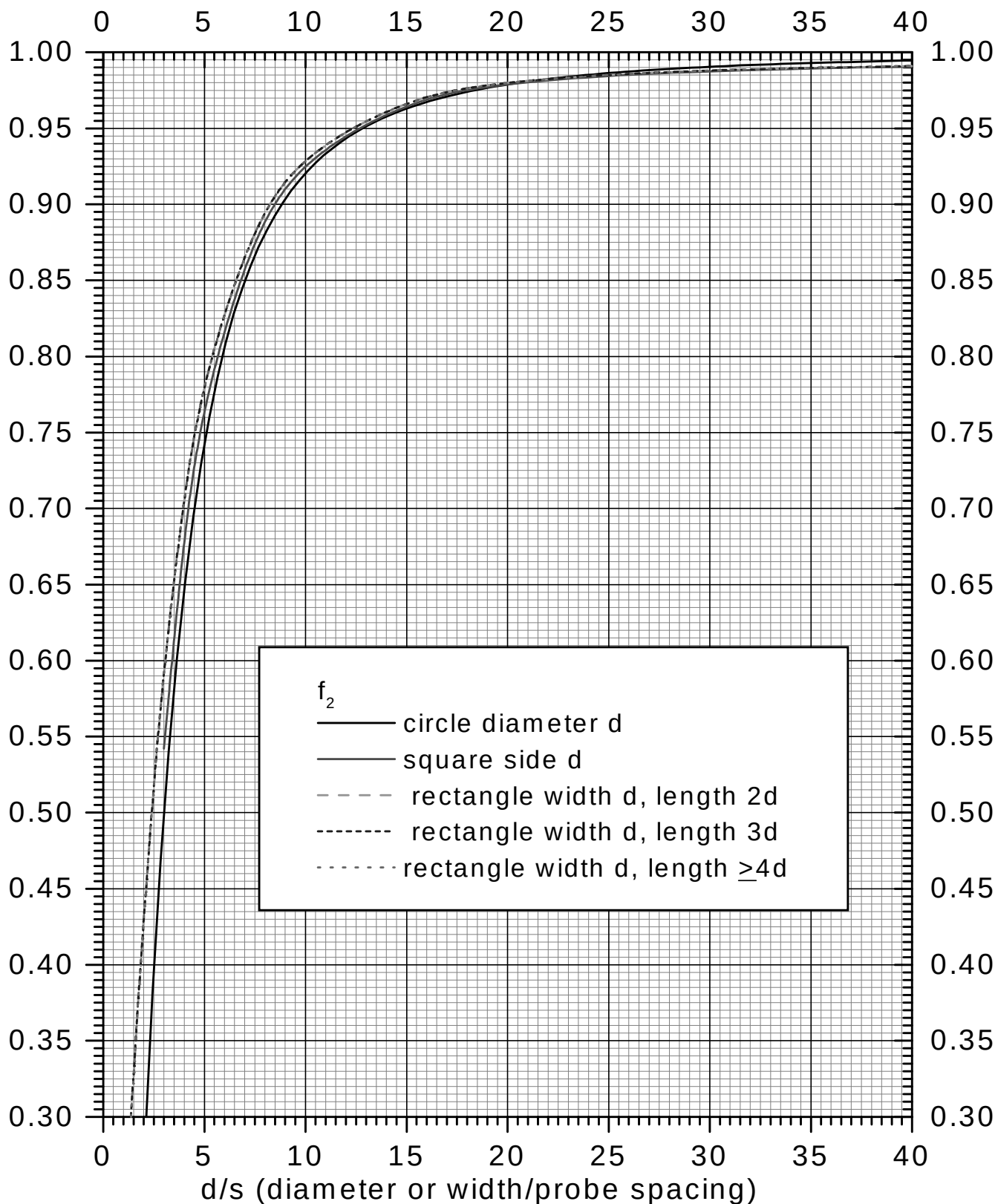
F.M.Smits, The Bell System Technical Journal **37**, 711-718 (1958)

More easily accessible reference:

D.K. Schroeder, *Semiconductor Material and Device Characterization*, Wiley 1990.



Source.: F.M.Smits, "Measurement of Sheet Resistivities with the Four-Point Probe", The Bell System Technical Journal 37, 711-718 (1958)



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